

UST-ORDYNSKY, Russian Federation

WMO: 307130

Lat: 52.817N

Lon: 104.767E

Elev: 526

StdP: 95.17

Time Zone: 8.00 (E08)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.9	-33.3	-37.9	0.1	-35.8	-35.4	0.1	-33.2	9.3	-19.0	7.9	-17.2	1.9	20	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	28.3	18.6	26.3	17.8	24.6	16.9	20.1	25.1	19.0	24.1	17.9	23.1	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	14.2	22.0	17.2	13.1	21.0	16.1	12.2	20.2	60.2	25.0	56.4	24.0	53.0	23.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.2	6.8	-39.2	32.1	2.9	1.8	-41.3	33.4	-43.0	34.4	-44.6	35.4	-46.7	36.8		
(5)			-39.1	22.0	2.8	1.5	-41.1	23.1	-42.7	23.9	-44.3	24.7	-46.3	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-22.4	-19.9	-10.0	0.3	8.6	14.5	17.4	15.3	8.4	0.4	-11.8	-19.8	(6)	
(7)		DBStd	14.91	5.55	5.51	6.19	4.19	4.21	3.87	3.00	2.87	3.82	4.26	6.46	6.26	(7)	
(8)		HDD10.0	4786	1005	836	621	291	77	7	1	1	74	297	654	923	(8)	
(9)		HDD18.3	7272	1263	1069	879	541	302	124	53	101	298	556	904	1182	(9)	
(10)		CDD10.0	597	0	0	0	0	34	142	229	166	26	0	0	0	(10)	
(11)		CDD18.3	41	0	0	0	0	1	9	23	8	0	0	0	0	(11)	
(12)		CDH23.3	637	0	0	0	0	37	206	281	103	9	0	0	0	(12)	
(13)		CDH26.7	135	0	0	0	0	4	61	52	18	0	0	0	0	(13)	
(14)	Wind	WSAvg	2.8	2.5	2.5	2.8	3.8	3.8	3.1	2.6	2.5	2.8	2.7	2.6	2.5	(14)	
(15)	Precipitation	PrecAvg	391	11	7	7	15	30	61	88	41	15	15	13		(15)	
(16)		PrecMax	598	23	15	24	45	57	152	200	193	110	36	30	31	(16)	
(17)		PrecMin	290	3	2	1	4	11	20	30	33	12	4	4	2	(17)	
(18)		PrecStd	72	6	3	5	12	13	30	36	42	23	7	7	7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	-2.0	11.6	17.2	27.3	31.4	30.0	24.9	16.3	5.0	-3.4		(19)	
(20)			MCWB	-8.5	-3.9	5.3	8.7	15.4	18.3	19.8	19.8	16.3	9.3	1.8	-4.1		(20)
(21)		2%	DB	-10.3	-7.0	4.4	13.6	24.4	28.4	28.2	26.2	21.0	12.8	1.2	-6.7		(21)
(22)			MCWB	-10.8	-7.7	1.0	6.4	13.4	17.8	19.3	18.6	14.1	7.4	-0.8	-7.3		(22)
(23)		5%	DB	-12.6	-9.4	1.9	10.9	21.6	25.6	26.4	24.0	18.1	9.9	-1.1	-8.9		(23)
(24)			MCWB	-13.1	-10.0	-0.5	4.6	12.0	16.5	18.6	17.5	12.0	5.9	-2.5	-9.4		(24)
(25)		10%	DB	-14.6	-11.5	-0.4	8.3	19.0	23.3	24.7	22.1	15.6	7.2	-2.9	-10.7		(25)
(26)			MCWB	-14.9	-12.1	-2.2	3.1	10.5	15.2	18.0	16.5	10.8	4.2	-3.8	-11.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.4	-3.7	5.5	9.0	16.0	20.6	21.8	21.3	16.3	9.8	1.9	-3.9		(27)
(28)			MCDWB	-7.7	-2.3	11.8	15.6	26.4	26.4	27.2	27.2	23.1	15.2	4.2	-3.0		(28)
(29)		2%	WB	-10.9	-7.8	1.4	6.9	14.1	18.7	20.5	19.4	14.5	7.9	-0.5	-7.4		(29)
(30)			MCDWB	-10.3	-6.9	3.9	13.5	23.1	25.2	25.6	24.4	19.7	12.3	1.1	-6.9		(30)
(31)		5%	WB	-13.0	-10.0	-0.4	5.0	12.5	17.5	19.5	18.1	12.8	6.2	-2.3	-9.3		(31)
(32)			MCDWB	-12.7	-9.3	1.9	10.5	20.6	24.0	24.3	23.2	17.1	9.3	-1.2	-8.9		(32)
(33)		10%	WB	-14.9	-12.0	-2.1	3.4	10.7	16.3	18.5	17.0	11.4	4.5	-3.9	-11.1		(33)
(34)			MCDWB	-14.6	-11.4	-0.3	7.9	18.1	22.1	23.3	21.1	14.7	7.1	-2.8	-10.7		(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	11.0	13.0	11.7	14.2	12.5	11.2	10.8	11.5	11.5	10.1	8.7	(35)	
(36)			MCDBR	9.8	11.7	12.9	16.7	20.2	18.5	15.5	15.2	16.7	16.0	11.6	9.5	(36)	
(37)			MCWBR	9.4	11.0	9.5	10.4	11.0	9.0	7.6	8.3	9.4	10.5	9.8	9.0	(37)	
(38)		5% WB	MCDWB	9.9	11.8	12.5	15.6	18.9	14.3	13.3	13.8	14.7	15.1	11.3	9.6	(38)	
(39)			MCWBR	9.4	11.0	9.5	10.1	10.5	7.8	7.4	7.8	8.7	10.3	9.7	9.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.272	0.297	0.329	0.375	0.432	0.439	0.427	0.387	0.338	0.315	0.294	0.274	(40)	
(41)		taud		2.083	2.070	2.103	2.223	2.086	2.171	2.248	2.345	2.438	2.400	2.211	2.072	(41)	
(42)		Ebn at Noon		687	790	846	855	817	816	819	836	843	782	663	601	(42)	
(43)		Edh at Noon		73	102	122	125	151	140	128	110	88	74	65	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.25	2.38	3.94	4.95	5.55	5.88	5.42	4.66	3.44	2.17	1.23	0.86	(44)	
(45)		RadStd		0.06	0.08	0.15	0.30	0.34	0.33	0.43	0.20	0.31	0.13	0.07	0.07	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page